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Professional Profile

Mathematical civil engineer with a master's degree in applied mathematics. Responsible, able to work in a team environment, with leadership skills, focused on meeting goals and objectives and always willing to improve and learn new things.

Education

Mar 2024 – Jul 2025	Universidad de Chile. Master's degree in applied mathematics. Graduate.	<i>Santiago, Chile</i>
Mar 2017 – Mar 2024	Universidad de Chile. Mathematical civil engineering. Graduate.	<i>Santiago, Chile</i>
Mar 2017 – Dec 2023	Universidad de Chile. Bachelor's degree in engineering science, major in mathematics. Graduate	<i>Santiago, Chile</i>

Professional Experience

Sept 2025 – Present	Centro de Modelamiento Matemático (CMM) Topic: Education (CMMedu). Position: Service provider. Function: - Designing and developing technology-enhanced mathematics learning experiences, including adaptive learning units, self-paced digital resources, and the mathematical content for a statistical analysis course for secondary school teachers, in collaboration with the Chilean Ministry of Education and FCFM, Universidad de Chile. - Developing activities for the first two units of an adaptive digital learning pathway as part of an online education pilot program funded through FONDEF Project 25110156.	<i>Santiago, Chile</i>
Jan – Apr 2025	Institut National de Recherche en Informatique et en Automatique (INRIA) Research Internship. Topic: Strategic Decision-Making in Decentralized Energy Systems. Position: Researcher Trainee. Function: Modeling decentralized energy systems through a strategic, non-cooperative lens. It explored how uncertainty and interdependence shape behavior in dynamic, non-cooperative environments.	<i>Lille, France</i>
Aug 2023 – Jan 2024	Centro Nacional de Inteligencia Artificial (CENIA). Internship III. Topic: Multiscale entropy in random graphs. Position: Research Assistant. Function: Studying and modeling entropy in random graphs and implementing its geometric multiscale.	<i>Santiago, Chile</i>
Dec 2021 – Feb 2022	ComDer Contraparte Central. Internship II. Topic: Modeling of financial valuation adjustments (XVA). Position: Trainee. Function: Studying and modeling three types of financial valuation adjustments (XVA) using Python.	<i>Santiago, Chile</i>

Jan – Feb 2021	Department of Mathematical Engineering, FCFM. Internship I. Topic: Satellite data study, InSAR satellites. Position: Research Assistant. Function: Searching for information required by supervisor.	Santiago, Chile
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Academic Experience

Aug 2025 – Present	Lecturer – FCFM, Universidad de Chile. Challenges of Innovation in Engineering and Sciences (CD1100) and Innovation Project in Engineering and Sciences (CD1201) - Wildfire mitigation: Using satellite data to protect ecosystems (Spring 2025). - Engineering in action: Protecting biodiversity (Fall 2026).	Santiago, Chile
Mar 2022 – Dec 2025	Teaching and Course Coordinator - FCFM - Universidad de Chile. - Coordinate teaching teams and course implementation for common-core mathematics and engineering innovation courses. - Support the planning of learning activities, assessment criteria, teaching materials, and coordination processes for large undergraduate courses. - Design and deliver training workshops for teaching assistants involved in assessment and feedback processes.	Santiago, Chile
Aug 2018 - Dec 2024	Teaching Assistant - FCFM - Universidad de Chile. - Conduct tutorials in Multivariable Calculus, Ordinary Differential Equations, Probability and Statistics, Discrete Mathematics, Theory of Computation, and engineering innovation courses. - Support assessment processes, student feedback, and the development of instructional materials.	Santiago, Chile

Extracurricular Activities

Dec 2017 - Dec 2023	University Outreach Assistant - Universidad de Chile. Support outreach and undergraduate admissions activities as part of the Unidad de Difusión para la Admisión de Pregrado (UDAP).	Santiago, Chile
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Skills

Languages	[Spanish]	Native
	[English]	Upper-Intermediate (B2)
	[French]	Basic level (currently learning)
Technical skills	Programming	Python, MATLAB, Julia, C, SQL, Java (basic)
	Data and Machine Learning	pandas, scikit-learn, TensorFlow/Keras, data visualization, statistical análisis
	Development and Tools	Streamlit, Git, GitHub, LaTeX, Docker (basic)
	Productivity and Analysis	Excel